

Chapter 2: Probability

STAT2920-2: Introduction to Probability

Mohamed Belalia, PhD.

University of Windsor
Department of Mathematics and Statistics



University
of Windsor

- 1 Introduction
- 2 Review of Set Theory
- 3 The Probability of an Event
- 4 Some Rules of Probability
- 5 Conditional Probability
- 6 Independent Events
- 7 Bayes' Theorem
- 8 Homework

Whole image

- Probability aims to formalize everyday sentences of the type:

“The chance of E is p ”,

where E is some **event** and p is some **measure** of the likelihood of occurrence of that event.

Whole image

- Probability aims to formalize everyday sentences of the type:

“The chance of E is p ”,

where E is some **event** and p is some **measure** of the likelihood of occurrence of that event.

Example 1.

- There is **35%** chance of rain,
- Due to *Covid-19*, there is **50%** chance that the winter term will be online too.
- I believe with a high probability the Liberal Party will win the upcoming election .
- The probability of success **STAT2920** course is relatively high.

- Historically, the oldest way of defining probabilities, the **classical probability concept**, applies when all possible outcomes are **equally likely**, as is presumably the case in most games of chance.
- We can then say that if there are N equally likely possibilities, of which one must occur and n are regarded as favorable, or as a **success**, then the probability of a **success** is given by the ratio $\frac{n}{N}$.
- Equivalently,

$$\mathbb{P}(\text{success}) = \frac{\# \text{ success}}{\# \text{ possible}} = \frac{n}{N}.$$

Example 2.

What is the probability of drawing an ace from an ordinary deck of 52 playing cards?

Example 2.

What is the probability of drawing an ace from an ordinary deck of 52 playing cards?

Solution:

- Since there are $n = 4$ aces among the $N = 52$ cards, the probability of drawing an ace is $\frac{4}{52} = \frac{1}{13}$.
- (It is assumed, of course, that each card has the same chance of being drawn.)

- Although equally likely possibilities are found mostly in games of chance,
- the classical probability concept applies also in a great variety of situations:
 - where gambling devices are used to make random selections
 - when office space is assigned to teaching assistants by lot
 - when some of the families in a township are chosen in such a way that each one has the same chance of being included in a sample study,
 - when machine parts are chosen for inspection so that each part produced has the same chance of being selected, and so forth.

- A major shortcoming of the classical probability concept is its limited applicability, for there are many situations in which the possibilities that arise cannot all be regarded as equally likely.
- This would be the case, for instance, if we are concerned with the question whether it will rain on a given day, if we are concerned with the outcome of an election, or if we are concerned with a person's recovery from a disease.
- Among the various probability concepts, most widely held is the frequency interpretation, according to which the probability of an event (outcome or happening) is the proportion of the time that events of the same kind will occur in the long run.

- If we say that the probability is 0.84 that a jet from Los Angeles to San Francisco will arrive on time, we mean (in accordance with the frequency interpretation) that such flights arrive on time 84 percent of the time.
- Similarly, if the weather bureau predicts that there is a 30 percent chance for rain (that is, a probability of 0.30), this means that under the same weather conditions it will rain 30 percent of the time.
- More generally, we say that an event has a probability of, say, 0.90, in the same sense in which we might say that our car will start in cold weather 90 percent of the time.

- We cannot guarantee what will happen on any particular occasion:
 - the car may start and then it may not,
 - but if we kept records over a long period of time, we should find that the proportion of “successes” is very close to 0.90.
- The approach to probability that we shall use in this chapter is the axiomatic approach, in which probabilities are defined as “mathematical objects” that behave according to certain well -defined rules.
- Then, any one of the preceding probability concepts, or interpretations, can be used in applications as long as it is consistent with these rules.

Definition 1 (Set).

A **set** is an unordered collection of objects called **elements** of the set.

Example 3.

- $\mathcal{A} = \{\clubsuit, \diamond, \heartsuit\}$
- $\mathcal{B} = \{5, 7, 9\};$
- $\mathcal{C} = \{x^3 : x = 1, 2, 3\} = \{1, 8, 27\};$
- $\mathcal{D} = \{H, T\}.$
- $\mathcal{E} = \{2, 3, 5, 7, 11, 13, 17, 19, \dots\}$

Definition 1 (Set).

A **set** is an unordered collection of objects called **elements** of the set.

Example 3.

- $\mathcal{A} = \{\clubsuit, \diamond, \heartsuit\}$
- $\mathcal{B} = \{5, 7, 9\};$
- $\mathcal{C} = \{x^3 : x = 1, 2, 3\} = \{1, 8, 27\};$
- $\mathcal{D} = \{H, T\}$. This is a **finite set**.
It has a finite number of elements.
- $\mathcal{E} = \{2, 3, 5, 7, 11, 13, 17, 19, \dots\}$

Definition 1 (Set).

A **set** is an unordered collection of objects called **elements** of the set.

Example 3.

- $\mathcal{A} = \{\clubsuit, \diamond, \heartsuit\}$
- $\mathcal{B} = \{5, 7, 9\};$
- $\mathcal{C} = \{x^3 : x = 1, 2, 3\} = \{1, 8, 27\};$
- $\mathcal{D} = \{H, T\}$. This is a **finite** set.
It has a finite number of elements.
- $\mathcal{E} = \{2, 3, 5, 7, 11, 13, 17, 19, \dots\}$ This is an **infinite** set. It has an infinite number of elements.

Some special sets:

- The set of natural numbers, $\mathbb{N} = \{1, 2, 3, \dots\}$.
- The set of integers, $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$.
- The set of real number $\mathbb{R}$.
- $\mathbb{Q} = \{\frac{a}{b} | a, b \in \mathbb{Z}, b \neq 0\}$.

Definition 2.

Two sets are **equal** if they contain exactly the same elements.

Definition 2.

Two sets are **equal** if they contain exactly the same elements.

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$D = \{20, 15, 10, 5\} \quad E = \{5, 10, 15, 20, \dots\}$$

Definition 2.

Two sets are **equal** if they contain exactly the same elements.

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$D = \{20, 15, 10, 5\} \quad E = \{5, 10, 15, 20, \dots\}$$

Definition 2.

Two sets are **equal** if they contain exactly the same elements.

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$D = \{20, 15, 10, 5\} \quad E = \{5, 10, 15, 20, \dots\}$$

“element of” symbol

$a \in \mathcal{A}$ means the element a belongs to the set $\mathcal{A}$.

$a \notin \mathcal{A}$ means the element a does not belong to the set $\mathcal{A}$.

$$\mathcal{A} = \{\clubsuit, \diamond\} \quad \diamond \in \mathcal{A}; \quad \heartsuit \notin \mathcal{A}$$

Definition 2.

Two sets are **equal** if they contain exactly the same elements.

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$D = \{20, 15, 10, 5\} \quad E = \{5, 10, 15, 20, \dots\}$$

“element of” symbol

$a \in \mathcal{A}$ means the element a belongs to the set $\mathcal{A}$.

$a \notin \mathcal{A}$ means the element a does not belong to the set $\mathcal{A}$.

$$\mathcal{A} = \{\clubsuit, \diamond\} \quad \diamond \in \mathcal{A}; \quad \heartsuit \notin \mathcal{A}$$

How can you complete these?

- $0 \in \square$
- $7 \square A$
- $35, 45, 55 \in \square$

Definition 2.

Two sets are **equal** if they contain exactly the same elements.

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$D = \{20, 15, 10, 5\} \quad E = \{5, 10, 15, 20, \dots\}$$

“element of” symbol

$a \in \mathcal{A}$ means the element a belongs to the set $\mathcal{A}$.

$a \notin \mathcal{A}$ means the element a does not belong to the set $\mathcal{A}$.

$$\mathcal{A} = \{\clubsuit, \diamond\} \quad \diamond \in \mathcal{A}; \quad \heartsuit \notin \mathcal{A}$$

How can you complete these?

- $0 \in \boxed{B}$
- $7 \boxed{} A$
- $35, 45, 55 \in \boxed{}$

Definition 2.

Two sets are **equal** if they contain exactly the same elements.

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$D = \{20, 15, 10, 5\} \quad E = \{5, 10, 15, 20, \dots\}$$

“element of” symbol

$a \in \mathcal{A}$ means the element a belongs to the set $\mathcal{A}$.

$a \notin \mathcal{A}$ means the element a does not belong to the set $\mathcal{A}$.

$$\mathcal{A} = \{\clubsuit, \diamond\} \quad \diamond \in \mathcal{A}; \quad \heartsuit \notin \mathcal{A}$$

How can you complete these?

- $0 \in \boxed{B}$
- $7 \boxed{\notin} A$
- $35, 45, 55 \in \boxed{}$

Definition 2.

Two sets are **equal** if they contain exactly the same elements.

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$D = \{20, 15, 10, 5\} \quad E = \{5, 10, 15, 20, \dots\}$$

“element of” symbol

$a \in \mathcal{A}$ means the element a belongs to the set $\mathcal{A}$.

$a \notin \mathcal{A}$ means the element a does not belong to the set $\mathcal{A}$.

$$\mathcal{A} = \{\clubsuit, \diamond\} \quad \diamond \in \mathcal{A}; \quad \heartsuit \notin \mathcal{A}$$

How can you complete these?

- $0 \in \boxed{B}$
- $7 \boxed{\notin} A$
- $35, 45, 55 \in \boxed{E}$

Definition 2.

Two sets are **equal** if they contain exactly the same elements.

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$D = \{20, 15, 10, 5\} \quad E = \{5, 10, 15, 20, \dots\}$$

“element of” symbol

$a \in \mathcal{A}$ means the element a belongs to the set $\mathcal{A}$.

$a \notin \mathcal{A}$ means the element a does not belong to the set $\mathcal{A}$.

$$\mathcal{A} = \{\clubsuit, \diamond\} \quad \diamond \in \mathcal{A}; \quad \heartsuit \notin \mathcal{A}$$

How can you complete these?

- $0 \in \boxed{B}$
- $7 \boxed{\notin} A$
- $35, 45, 55 \in \boxed{E}$ This is a common “abuse of notation”

Definition 2.

Two sets are **equal** if they contain exactly the same elements.

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$D = \{20, 15, 10, 5\} \quad E = \{5, 10, 15, 20, \dots\}$$

"element of" symbol

$a \in \mathcal{A}$ means the element a belongs to the set $\mathcal{A}$.

$a \notin \mathcal{A}$ means the element a does not belong to the set $\mathcal{A}$.

$$\mathcal{A} = \{\clubsuit, \diamond\} \quad \diamond \in \mathcal{A}; \quad \heartsuit \notin \mathcal{A}$$

We write $|A|$ to mean the size, or **cardinality** of a set A . For finite sets, this means the number of distinct elements.

Definition 2.

Two sets are **equal** if they contain exactly the same elements.

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$D = \{20, 15, 10, 5\} \quad E = \{5, 10, 15, 20, \dots\}$$

“element of” symbol

$a \in \mathcal{A}$ means the element a belongs to the set $\mathcal{A}$.

$a \notin \mathcal{A}$ means the element a does not belong to the set $\mathcal{A}$.

$$\mathcal{A} = \{\clubsuit, \diamond\} \quad \diamond \in \mathcal{A}; \quad \heartsuit \notin \mathcal{A}$$

We write $|A|$ to mean the size, or **cardinality** of a set A . For finite sets, this means the number of distinct elements.

How can you complete these?

- $|A| = \square$
- $|A| = \left| \square \right|$

Definition 2.

Two sets are **equal** if they contain exactly the same elements.

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$D = \{20, 15, 10, 5\} \quad E = \{5, 10, 15, 20, \dots\}$$

“element of” symbol

$a \in \mathcal{A}$ means the element a belongs to the set $\mathcal{A}$.

$a \notin \mathcal{A}$ means the element a does not belong to the set $\mathcal{A}$.

$$\mathcal{A} = \{\clubsuit, \diamond\} \quad \diamond \in \mathcal{A}; \quad \heartsuit \notin \mathcal{A}$$

We write $|A|$ to mean the size, or **cardinality** of a set A . For finite sets, this means the number of distinct elements.

How can you complete these?

- $|A| = \boxed{4}$
- $|A| = \boxed{}$

Definition 2.

Two sets are **equal** if they contain exactly the same elements.

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$D = \{20, 15, 10, 5\} \quad E = \{5, 10, 15, 20, \dots\}$$

“element of” symbol

$a \in \mathcal{A}$ means the element a belongs to the set $\mathcal{A}$.

$a \notin \mathcal{A}$ means the element a does not belong to the set $\mathcal{A}$.

$$\mathcal{A} = \{\clubsuit, \diamond\} \quad \diamond \in \mathcal{A}; \quad \heartsuit \notin \mathcal{A}$$

We write $|A|$ to mean the size, or **cardinality** of a set A . For finite sets, this means the number of distinct elements.

How can you complete these?

- $|A| = \boxed{4}$
- $|A| = \boxed{D}$

Definition 2.

Two sets are **equal** if they contain exactly the same elements.

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$D = \{20, 15, 10, 5\} \quad E = \{5, 10, 15, 20, \dots\}$$

“element of” symbol

$a \in \mathcal{A}$ means the element a belongs to the set $\mathcal{A}$.

$a \notin \mathcal{A}$ means the element a does not belong to the set $\mathcal{A}$.

$$\mathcal{A} = \{\clubsuit, \diamond\} \quad \diamond \in \mathcal{A}; \quad \heartsuit \notin \mathcal{A}$$

We write $|A|$ to mean the size, or **cardinality** of a set A . For finite sets, this means the number of distinct elements.

- True or false: $C=3$
- True or false: If two sets are equal, their cardinalities are equal.

Definition 2.

Two sets are **equal** if they contain exactly the same elements.

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$D = \{20, 15, 10, 5\} \quad E = \{5, 10, 15, 20, \dots\}$$

“element of” symbol

$a \in \mathcal{A}$ means the element a belongs to the set $\mathcal{A}$.

$a \notin \mathcal{A}$ means the element a does not belong to the set $\mathcal{A}$.

$$\mathcal{A} = \{\clubsuit, \diamond\} \quad \diamond \in \mathcal{A}; \quad \heartsuit \notin \mathcal{A}$$

We write $|A|$ to mean the size, or **cardinality** of a set A . For finite sets, this means the number of distinct elements.

- True or false: $C=3$ False. C is a set, 3 is a number.
- True or false: If two sets are equal, their cardinalities are equal.

Definition 2.

Two sets are **equal** if they contain exactly the same elements.

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$D = \{20, 15, 10, 5\} \quad E = \{5, 10, 15, 20, \dots\}$$

“element of” symbol

$a \in \mathcal{A}$ means the element a belongs to the set $\mathcal{A}$.

$a \notin \mathcal{A}$ means the element a does not belong to the set $\mathcal{A}$.

$$\mathcal{A} = \{\clubsuit, \diamond\} \quad \diamond \in \mathcal{A}; \quad \heartsuit \notin \mathcal{A}$$

We write $|A|$ to mean the size, or **cardinality** of a set A . For finite sets, this means the number of distinct elements.

- **True or false:** $C=3$ False. C is a set, 3 is a number.
- **True or false:** If two sets are equal, their cardinalities are equal.

True.

Definition 3.

A set A is a **subset** of a set B if every element of A is also an element of B . We write

$$A \subseteq B$$

Definition 3.

A set A is a **subset** of a set B if every element of A is also an element of B . We write

$$A \subseteq B$$

If A is not a subset of B , we write

$$A \not\subseteq B$$

Definition 3.

A set A is a **subset** of a set B if every element of A is also an element of B . We write

$$A \subseteq B$$

If A is not a subset of B , we write

$$A \not\subseteq B$$

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

Definition 3.

A set A is a **subset** of a set B if every element of A is also an element of B . We write

$$A \subseteq B$$

If A is not a subset of B , we write

$$A \not\subseteq B$$

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$A \subseteq$$

Definition 3.

A set A is a **subset** of a set B if every element of A is also an element of B . We write

$$A \subseteq B$$

If A is not a subset of B , we write

$$A \not\subseteq B$$

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$A \subseteq B$$

Definition 3.

A set A is a **subset** of a set B if every element of A is also an element of B . We write

$$A \subseteq B$$

If A is not a subset of B , we write

$$A \not\subseteq B$$

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$A \subseteq B \quad C \subseteq B$$

Definition 3.

A set A is a **subset** of a set B if every element of A is also an element of B . We write

$$A \subseteq B$$

If A is not a subset of B , we write

$$A \not\subseteq B$$

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$A \subseteq B \quad C \subseteq A, B$$

Definition 3.

A set A is a **subset** of a set B if every element of A is also an element of B . We write

$$A \subseteq B$$

If A is not a subset of B , we write

$$A \not\subseteq B$$

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$A \subseteq B$$

$$C \subseteq A, B$$

True or false: $A \subseteq A$

Definition 3.

A set A is a **subset** of a set B if every element of A is also an element of B . We write

$$A \subseteq B$$

If A is not a subset of B , we write

$$A \not\subseteq B$$

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$A \subseteq B$$

$$C \subseteq A, B$$

True or false: $A \subseteq A$ True

Definition 3.

A set A is a **subset** of a set B if every element of A is also an element of B . We write

$$A \subseteq B$$

If A is not a subset of B , we write

$$A \not\subseteq B$$

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$A \subseteq B \quad C \subseteq A, B \quad \text{True or false: } A \subseteq A \quad \text{True}$$

True or false: if E and F are finite sets, and $E \subseteq F$, then $|E| < |F|$.

Definition 3.

A set A is a **subset** of a set B if every element of A is also an element of B . We write

$$A \subseteq B$$

If A is not a subset of B , we write

$$A \not\subseteq B$$

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$A \subseteq B \quad C \subseteq A, B \quad \text{True or false: } A \subseteq A \quad \text{True}$$

True or false: if E and F are finite sets, and $E \subseteq F$, then $|E| < |F|$.

If E and F are finite sets, $E \subseteq F$ and $|E| \geq |F|$, what do you know about the relationship between E and F ?

Definition 3.

A set A is a **subset** of a set B if every element of A is also an element of B . We write

$$A \subseteq B$$

If A is not a subset of B , we write

$$A \not\subseteq B$$

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$A \subseteq B \quad C \subseteq A, B \quad \text{True or false: } A \subseteq A \quad \text{True}$$

True or false: if E and F are finite sets, and $E \subseteq F$, then $|E| < |F|$.

If E and F are finite sets, $E \subseteq F$ and $|E| \geq |F|$, what do you know about the relationship between E and F ?

True or false: for any set E , $\emptyset \subseteq E$.

Definition 3.

A set A is a **subset** of a set B if every element of A is also an element of B . We write

$$A \subseteq B$$

If A is not a subset of B , we write

$$A \not\subseteq B$$

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$A \subseteq B \quad C \subseteq A, B \quad \text{True or false: } A \subseteq A \quad \text{True}$$

True or false: if E and F are finite sets, and $E \subseteq F$, then $|E| < |F|$.

If E and F are finite sets, $E \subseteq F$ and $|E| \geq |F|$, what do you know about the relationship between E and F ?

True or false: for any set E , $\emptyset \subseteq E$.

If $E \subseteq F$ and $F \subseteq E$, what do you know about E and F ?

Definition 3.

A set A is a **subset** of a set B if every element of A is also an element of B . We write

$$A \subseteq B$$

If A is not a subset of B , we write

$$A \not\subseteq B$$

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$A \subseteq B \quad C \subseteq A, B \quad \text{True or false: } A \subseteq A \quad \text{True}$$

True or false: if E and F are finite sets, and $E \subseteq F$, then $|E| < |F|$.
False (see next question)

If E and F are finite sets, $E \subseteq F$ and $|E| \geq |F|$, what do you know about the relationship between E and F ?

True or false: for any set E , $\emptyset \subseteq E$.

If $E \subseteq F$ and $F \subseteq E$, what do you know about E and F ?

Definition 3.

A set A is a **subset** of a set B if every element of A is also an element of B . We write

$$A \subseteq B$$

If A is not a subset of B , we write

$$A \not\subseteq B$$

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$A \subseteq B \quad C \subseteq A, B \quad \text{True or false: } A \subseteq A \quad \text{True}$$

True or false: if E and F are finite sets, and $E \subseteq F$, then $|E| < |F|$.
False (see next question)

If E and F are finite sets, $E \subseteq F$ and $|E| \geq |F|$, what do you know about the relationship between E and F ? $E = F$ (and so $|E| = |F|$)

True or false: for any set E , $\emptyset \subseteq E$.

If $E \subseteq F$ and $F \subseteq E$, what do you know about E and F ?

Definition 3.

A set A is a **subset** of a set B if every element of A is also an element of B . We write

$$A \subseteq B$$

If A is not a subset of B , we write

$$A \not\subseteq B$$

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$A \subseteq B \quad C \subseteq A, B \quad \text{True or false: } A \subseteq A \quad \text{True}$$

True or false: if E and F are finite sets, and $E \subseteq F$, then $|E| < |F|$.
False (see next question)

If E and F are finite sets, $E \subseteq F$ and $|E| \geq |F|$, what do you know about the relationship between E and F ? $E = F$ (and so $|E| = |F|$)

True or false: for any set E , $\emptyset \subseteq E$. True

If $E \subseteq F$ and $F \subseteq E$, what do you know about E and F ?

Definition 3.

A set A is a **subset** of a set B if every element of A is also an element of B . We write

$$A \subseteq B$$

If A is not a subset of B , we write

$$A \not\subseteq B$$

$$A = \{5, 10, 15, 20\} \quad B = \{0, 5, 10, 15, 20\} \quad C = \{5, 10, 15\}$$

$$A \subseteq B$$

$$C \subseteq A, B$$

True or false: $A \subseteq A$ True

True or false: if E and F are finite sets, and $E \subseteq F$, then $|E| < |F|$.
False (see next question)

If E and F are finite sets, $E \subseteq F$ and $|E| \geq |F|$, what do you know about the relationship between E and F ? $E = F$ (and so $|E| = |F|$)

True or false: for any set E , $\emptyset \subseteq E$. True

If $E \subseteq F$ and $F \subseteq E$, what do you know about E and F ? $E = F$ (!!!)

True or False?

- $a \subseteq \{a, b, c\}$
- $a \in \{a, b, c\}$
- $\{a\} \subseteq \{a, b, c\}$
- $\{a\} \in \{a, b, c\}$
- $a \in \{\{a\}, \{b, c\}\}$
- $\{a\} \in \{\{a\}, \{b, c\}\}$
- $a \subseteq \{\{a\}, \{b, c\}\}$
- $\{a\} \subseteq \{\{a\}, \{b, c\}\}$
- $\{\{a\}\} \subseteq \{\{a\}, \{b, c\}\}$

True or False?

- $a \subseteq \{a, b, c\}$ False
- $a \in \{a, b, c\}$ True
- $\{a\} \subseteq \{a, b, c\}$ True
- $\{a\} \in \{a, b, c\}$ False
- $a \in \{\{a\}, \{b, c\}\}$ False
- $\{a\} \in \{\{a\}, \{b, c\}\}$ True
- $a \subseteq \{\{a\}, \{b, c\}\}$ False
- $\{a\} \subseteq \{\{a\}, \{b, c\}\}$ False
- $\{\{a\}\} \subseteq \{\{a\}, \{b, c\}\}$ True

Example: List all subsets of the set $S = \{a, b, c\}$.

Example: List all subsets of the set $S = \{a, b, c\}$.

size	sets	notes
0		
1		
2		
3		

Example: List all subsets of the set $S = \{a, b, c\}$.

size	sets	notes
0	$\emptyset$	subset of any set
1		
2		
3		

Example: List all subsets of the set $S = \{a, b, c\}$.

size	sets	notes
0	$\emptyset$	subset of any set
1	$\{a\}$ $\{b\}$ $\{c\}$	
2		
3		

Example: List all subsets of the set $S = \{a, b, c\}$.

size	sets	notes
0	$\emptyset$	subset of any set
1	$\{a\}$ $\{b\}$ $\{c\}$	
2		delete one element
3		

Example: List all subsets of the set $S = \{a, b, c\}$.

size	sets	notes
0	$\emptyset$	subset of any set
1	$\{a\}$ $\{b\}$ $\{c\}$	
2	$\{b, c\}$	delete one element
3		

Example: List all subsets of the set $S = \{a, b, c\}$.

size	sets	notes
0	$\emptyset$	subset of any set
1	$\{a\}$ $\{b\}$ $\{c\}$	
2	$\{b, c\}$ $\{a, c\}$	delete one element
3		

Example: List all subsets of the set $S = \{a, b, c\}$.

size	sets	notes
0	$\emptyset$	subset of any set
1	$\{a\}$ $\{b\}$ $\{c\}$	
2	$\{b, c\}$ $\{a, c\}$ $\{a, b\}$	delete one element
3		

Example: List all subsets of the set $S = \{a, b, c\}$.

size	sets	notes
0	$\emptyset$	subset of any set
1	$\{a\}$ $\{b\}$ $\{c\}$	
2	$\{b, c\}$ $\{a, c\}$ $\{a, b\}$	delete one element
3	$\{a, b, c\}$	

Example: List all subsets of the set $S = \{a, b, c\}$.

size	sets	notes
0	$\emptyset$	subset of any set
1	$\{a\}$ $\{b\}$ $\{c\}$	
2	$\{b, c\}$ $\{a, c\}$ $\{a, b\}$	delete one element
3	$\{a, b, c\}$	

Example: List all subsets of the set $T = \{a, b, \{1, 2\}\}$.

Example: List all subsets of the set $S = \{a, b, c\}$.

size	sets			notes
0	$\emptyset$			subset of any set
1	$\{a\}$	$\{b\}$	$\{c\}$	
2	$\{b, c\}$	$\{a, c\}$	$\{a, b\}$	delete one element
3	$\{a, b, c\}$			

Example: List all subsets of the set $T = \{a, b, \{1, 2\}\}$.

Note: $|T| = 3$.

Example: List all subsets of the set $S = \{a, b, c\}$.

size	sets	notes
0	$\emptyset$	subset of any set
1	$\{a\}$ $\{b\}$ $\{c\}$	
2	$\{b, c\}$ $\{a, c\}$ $\{a, b\}$	delete one element
3	$\{a, b, c\}$	

Example: List all subsets of the set $T = \{a, b, \{1, 2\}\}$.

Note: $|T| = 3$.

size	sets	notes
0		
1		
2		
3		

Example: List all subsets of the set $S = \{a, b, c\}$.

size	sets	notes
0	$\emptyset$	subset of any set
1	$\{a\}$ $\{b\}$ $\{c\}$	
2	$\{b, c\}$ $\{a, c\}$ $\{a, b\}$	delete one element
3	$\{a, b, c\}$	

Example: List all subsets of the set $T = \{a, b, \{1, 2\}\}$.

Note: $|T| = 3$.

size	sets	notes
0	$\emptyset$	subset of any set
1		
2		
3		

Example: List all subsets of the set $S = \{a, b, c\}$.

size	sets	notes
0	$\emptyset$	subset of any set
1	$\{a\}$ $\{b\}$ $\{c\}$	
2	$\{b, c\}$ $\{a, c\}$ $\{a, b\}$	delete one element
3	$\{a, b, c\}$	

Example: List all subsets of the set $T = \{a, b, \{1, 2\}\}$.

Note: $|T| = 3$.

size	sets	notes
0	$\emptyset$	subset of any set
1	$\{a\}$ $\{b\}$ $\{\{1, 2\}\}$	
2		
3		

Example: List all subsets of the set $S = \{a, b, c\}$.

size	sets	notes
0	$\emptyset$	subset of any set
1	$\{a\}$ $\{b\}$ $\{c\}$	
2	$\{b, c\}$ $\{a, c\}$ $\{a, b\}$	delete one element
3	$\{a, b, c\}$	

Example: List all subsets of the set $T = \{a, b, \{1, 2\}\}$.

Note: $|T| = 3$.

size	sets	notes
0	$\emptyset$	subset of any set
1	$\{a\}$ $\{b\}$ $\{\{1, 2\}\}$	
2	$\{b, \{1, 2\}\}$ $\{a, \{1, 2\}\}$ $\{a, b\}$	delete one element
3		

Example: List all subsets of the set $S = \{a, b, c\}$.

size	sets	notes
0	$\emptyset$	subset of any set
1	$\{a\}$ $\{b\}$ $\{c\}$	
2	$\{b, c\}$ $\{a, c\}$ $\{a, b\}$	delete one element
3	$\{a, b, c\}$	

Example: List all subsets of the set $T = \{a, b, \{1, 2\}\}$.

Note: $|T| = 3$.

size	sets	notes
0	$\emptyset$	subset of any set
1	$\{a\}$ $\{b\}$ $\{\{1, 2\}\}$	
2	$\{b, \{1, 2\}\}$ $\{a, \{1, 2\}\}$ $\{a, b\}$	delete one element
3	$\{a, b, \{1, 2\}\}$	

Example: List all subsets of the set $S = \{a, b, c\}$.

size	sets	notes
0	$\emptyset$	subset of any set
1	$\{a\}$ $\{b\}$ $\{c\}$	
2	$\{b, c\}$ $\{a, c\}$ $\{a, b\}$	delete one element
3	$\{a, b, c\}$	

Example: List all subsets of the set $T = \{a, b, \{1, 2\}\}$.

Note: $|T| = 3$.

size	sets	notes
0	$\emptyset$	subset of any set
1	$\{a\}$ $\{b\}$ $\{\{1, 2\}\}$	
2	$\{b, \{1, 2\}\}$ $\{a, \{1, 2\}\}$ $\{a, b\}$	delete one element
3	$\{a, b, \{1, 2\}\}$	

How many subsets does *any* set of cardinality 3 have?

Example: List all subsets of the set $S = \{a, b, c\}$.

size	sets	notes
0	$\emptyset$	subset of any set
1	$\{a\}$ $\{b\}$ $\{c\}$	
2	$\{b, c\}$ $\{a, c\}$ $\{a, b\}$	delete one element
3	$\{a, b, c\}$	

Example: List all subsets of the set $T = \{a, b, \{1, 2\}\}$.

Note: $|T| = 3$.

size	sets	notes
0	$\emptyset$	subset of any set
1	$\{a\}$ $\{b\}$ $\{\{1, 2\}\}$	
2	$\{b, \{1, 2\}\}$ $\{a, \{1, 2\}\}$ $\{a, b\}$	delete one element
3	$\{a, b, \{1, 2\}\}$	

How many subsets does *any* set of cardinality 3 have? 8

- $S = \Omega$ is the **universal** set i.e. the collection of all points of interest for the present situation.
 - Rolling a dice $\implies S = \{1, 2, 3, 4, 5, 6\}$
 - tossing of a coin once $\implies S = \{H, T\}$

- $S = \Omega$ is the **universal** set i.e. the collection of all points of interest for the present situation.
 - Rolling a dice $\implies S = \{1, 2, 3, 4, 5, 6\}$
 - tossing of a coin once $\implies S = \{H, T\}$
- the **Empty Set**, $\emptyset$. This is the set with no elements.

- $S = \Omega$ is the **universal** set i.e. the collection of all points of interest for the present situation.
 - Rolling a dice $\implies S = \{1, 2, 3, 4, 5, 6\}$
 - tossing of a coin once $\implies S = \{H, T\}$
- the **Empty Set**, $\emptyset$. This is the set with no elements.
 - $\emptyset = \{\}$

- $S = \Omega$ is the **universal** set i.e. the collection of all points of interest for the present situation.
 - Rolling a dice $\implies S = \{1, 2, 3, 4, 5, 6\}$
 - tossing of a coin once $\implies S = \{H, T\}$
- the **Empty Set**, $\emptyset$. This is the set with no elements.
 - $\emptyset = \{\}$
 - $|\emptyset| =$

- $S = \Omega$ is the **universal** set i.e. the collection of all points of interest for the present situation.
 - Rolling a dice $\implies S = \{1, 2, 3, 4, 5, 6\}$
 - tossing of a coin once $\implies S = \{H, T\}$
- the **Empty Set**, $\emptyset$. This is the set with no elements.
 - $\emptyset = \{\}$
 - $|\emptyset| = 0$

- $S = \Omega$ is the **universal** set i.e. the collection of all points of interest for the present situation.
 - Rolling a dice $\implies S = \{1, 2, 3, 4, 5, 6\}$
 - tossing of a coin once $\implies S = \{H, T\}$
- the **Empty Set**, $\emptyset$. This is the set with no elements.
 - $\emptyset = \{\}$
 - $|\emptyset| = 0$
 - **True or false:** If $|A| = 0$, then $A = \emptyset$.

- $S = \Omega$ is the **universal** set i.e. the collection of all points of interest for the present situation.
 - Rolling a dice $\implies S = \{1, 2, 3, 4, 5, 6\}$
 - tossing of a coin once $\implies S = \{H, T\}$
- the **Empty Set**, $\emptyset$. This is the set with no elements.
 - $\emptyset = \{\}$
 - $|\emptyset| = 0$
 - **True or false:** If $|A| = 0$, then $A = \emptyset$. True.

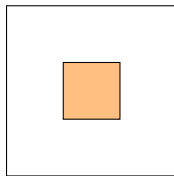
- $S = \Omega$ is the **universal** set i.e. the collection of all points of interest for the present situation.
 - Rolling a dice $\implies S = \{1, 2, 3, 4, 5, 6\}$
 - tossing of a coin once $\implies S = \{H, T\}$
- the **Empty Set**, $\emptyset$. This is the set with no elements.
 - $\emptyset = \{\}$
 - $|\emptyset| = 0$
 - True or false: If $|A| = 0$, then $A = \emptyset$. True.
 - True or false: $\emptyset = \{\emptyset\}$.

- $S = \Omega$ is the **universal** set i.e. the collection of all points of interest for the present situation.
 - Rolling a dice $\implies S = \{1, 2, 3, 4, 5, 6\}$
 - tossing of a coin once $\implies S = \{H, T\}$
- the **Empty Set**, $\emptyset$. This is the set with no elements.
 - $\emptyset = \{\}$
 - $|\emptyset| = 0$
 - **True or false:** If $|A| = 0$, then $A = \emptyset$. True.
 - **True or false:** $\emptyset = \{\emptyset\}$.
False. The set $\{\emptyset\}$ has one element: the empty set.

- $S = \Omega$ is the **universal** set i.e. the collection of all points of interest for the present situation.
 - Rolling a dice $\implies S = \{1, 2, 3, 4, 5, 6\}$
 - tossing of a coin once $\implies S = \{H, T\}$
- the **Empty Set**, $\emptyset$. This is the set with no elements.
 - $\emptyset = \{\}$
 - $|\emptyset| = 0$
 - True or false: If $|A| = 0$, then $A = \emptyset$. True.
 - True or false: $\emptyset = \{\emptyset\}$.
False. The set $\{\emptyset\}$ has one element: the empty set.
"An empty box is not the same as a box with an empty box inside it."



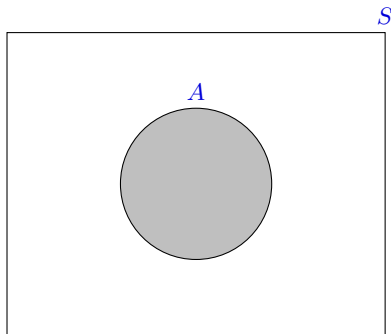
$\emptyset$



$\{\emptyset\}$

- **Venn diagrams** are very useful in visualizing relation between sets.
- In a Venn diagram any set is depicted by a closed region.

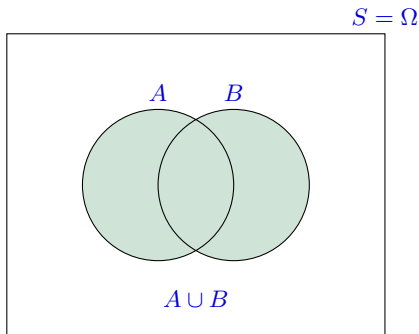
- **Venn diagrams** are very useful in visualizing relation between sets.
- In a Venn diagram any set is depicted by a closed region.



Definition 4.

The **union** of A and B is denoted $A \cup B$ (green area in venn diagram):

$$\omega \in A \cup B \Leftrightarrow \omega \in A \text{ or } \omega \in B \text{ or both}$$



Remark 1.

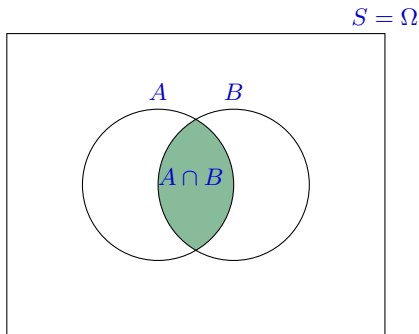
For all subsets A of S ,
 $A \cup \emptyset = A$ and $A \cup S = S$

Definition 5.

The **intersection** of A and B is denoted $A \cap B$ (green area in venn diagram):

$$\omega \in A \cap B \Leftrightarrow \omega \in A \text{ and } \omega \in B$$

A and B are disjoint if $A \cap B = \emptyset$



Remark 2.

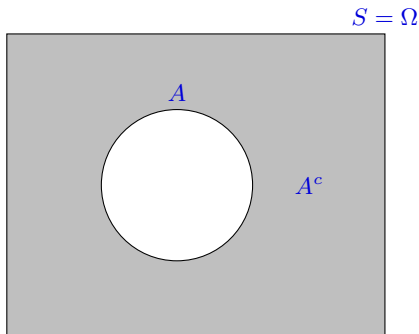
For all subsets A of S ,
 $A \cap \emptyset = \emptyset$ and $A \cap S = A$

Definition 6.

The **complement** of $A \subset S$ is denoted $A^c = \overline{A} \subset S$ (gray area in venn diagram):

$$\omega \in A^c \Leftrightarrow \omega \notin A.$$

It comes out that $(A^c)^c = A$.

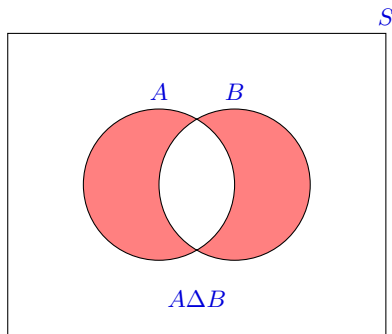
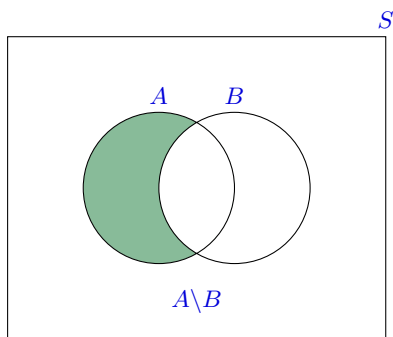


Remark 3.

the complement of S is the empty set $\emptyset$: $S^c = \emptyset$

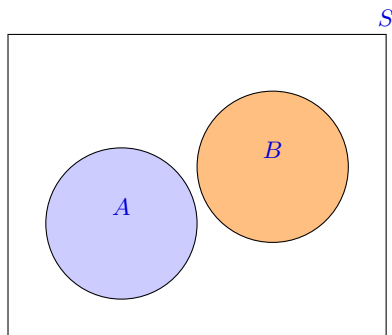
Definition 7.

The **difference** of $A - B = A \setminus B = A \cap B^c$ and the **symmetric difference** $A \Delta B = (A \setminus B) \cup (B \setminus A)$



Definition 8.

Two sets having no elements in common are said to be **mutually exclusive**.



- Let, $S = \{1, 2, 3, 4, 5\}$, $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$ and $C = \{4\}$
- then $A \cup B = \{1, 2, 3, 4\}$
- $A \cap B = \{2, 3\}$
- $A \cap C = \emptyset$
- $C \subset B$ thus $B \cup C = B$ and $B \cap C = C$
- $A^c = \{4, 5\}$

- Let, $S = \{Red, Blue, Green, Yellow, Pink\}$, $A = \{Red, Blue, Green\}$, $B = \{Red, Green, Yellow\}$ and $C = \{Blue, Pink\}$
- $A \cup B = \dots\dots\dots$
- $A \cap B = \dots$
- $A \cap C = \dots$
- $A^c = \dots$

Law 1 (Commutative Laws).

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

Law 2 (Associative Laws).

$$(A \cup B) \cup C = A \cup (B \cup C)$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

Law 3 (Distributive Laws).

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$$

$$(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$$

Law 4.

$$A \cap A^c = \emptyset$$

$$A \cup A^c = S$$

Law 5 (De Morgan's Laws).



$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

- *Generalization*

$$(A_1 \cup \cdots \cup A_n)^c = A_1^c \cap \cdots \cap A_n^c$$

$$(A_1 \cap \cdots \cap A_n)^c = A_1^c \cup \cdots \cup A_n^c$$

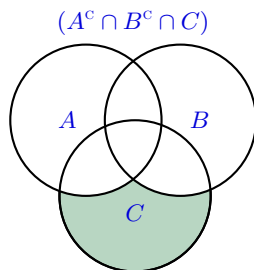
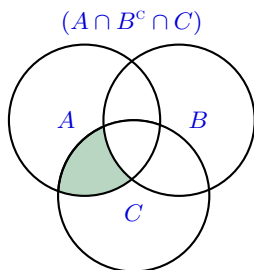
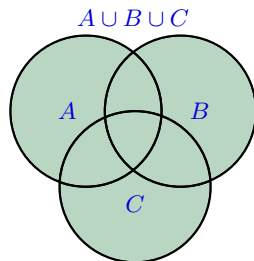
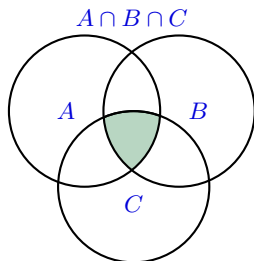
Proof.

$$\begin{aligned}\omega &\in (A_1 \cup \dots \cup A_n)^c \\ &\iff \omega \notin A_1 \cup \dots \cup A_n \\ &\iff \omega \notin A_1 \text{ and } \dots \text{ and } \omega \notin A_n \\ &\iff \omega \in A_1^c \text{ and } \dots \text{ and } \omega \in A_n^c \\ &\iff \omega \in A_1^c \cap \dots \cap A_n^c\end{aligned}$$

Similarly,

$$\begin{aligned}\omega &\in (A_1 \cap \dots \cap A_n)^c \\ &\iff \omega \notin A_1 \cap \dots \cap A_n \\ &\iff \omega \notin A_1 \text{ or } \dots \text{ or } \omega \notin A_n \\ &\iff \omega \in A_1^c \text{ or } \dots \text{ or } \omega \in A_n^c \\ &\iff \omega \in A_1^c \cup \dots \cup A_n^c\end{aligned}$$





Definition 9 (Axioms).

Let S be the sample space

1. **Axiom 1:** The probability of an event is a nonnegative real number; that is, $\mathbb{P}(A) \geq 0$ for any subset A of S .
2. **Axiom 2:** $\mathbb{P}(S) = 1$.
3. **Axiom 3:** If $A_1, A_2, A_3, \dots$, is a finite or infinite sequence of mutually exclusive events of S , then

$$\mathbb{P}(A_1 \cup A_2 \cup A_3 \cup \dots) = \mathbb{P}(A_1) + \mathbb{P}(A_2) + \mathbb{P}(A_3) + \dots$$

$$\iff \mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$$

Example 4.

Suppose an experiment consists of four mutually exclusive outcomes (events): A , B , C , and D . Why the following probabilities are wrong?

a. $\mathbb{P}(A) = 0.12, \mathbb{P}(B) = 0.63, \mathbb{P}(C) = 0.45$ and $\mathbb{P}(D) = -0.2$.

b. $\mathbb{P}(A) = \frac{9}{120}, \mathbb{P}(B) = \frac{45}{120}, \mathbb{P}(C) = \frac{27}{120}$ and $\mathbb{P}(D) = \frac{46}{120}$.

Example 4.

Suppose an experiment consists of four mutually exclusive outcomes (events): A, B, C , and D . Why the following probabilities are wrong?

a. $\mathbb{P}(A) = 0.12, \mathbb{P}(B) = 0.63, \mathbb{P}(C) = 0.45$ and $\mathbb{P}(D) = -0.2$.

b. $\mathbb{P}(A) = \frac{9}{120}, \mathbb{P}(B) = \frac{45}{120}, \mathbb{P}(C) = \frac{27}{120}$ and $\mathbb{P}(D) = \frac{46}{120}$.

Solution:

a. $\mathbb{P}(D) = -0.2$ violates Axiom 1;

b. $P(S) = P(A \cup B \cup C) = \frac{9}{120} + \frac{45}{120} + \frac{27}{120} + \frac{46}{120} = \frac{127}{120} \neq 1$, and violates Axiom 2.

Theorem 1.

If A is an event in a discrete sample space S , then $\mathbb{P}(A)$ equals the sum of the probabilities of the individual outcomes comprising A .

Proof of Theorem 1.

Let $O_1, O_2, O_3, \dots$, be the finite or infinite sequence of outcomes that comprise the event A . Thus,

$$A = O_1 \cup O_2 \cup O_3 \dots$$

and since the individual outcomes, the O 's, are mutually exclusive, the third postulate of probability yields

$$\mathbb{P}(A) = \mathbb{P}(O_1) + \mathbb{P}(O_2) + \mathbb{P}(O_3) + \dots$$

This completes the proof. □

Example 5.

If we twice flip a balanced coin, what is the probability of getting at least one head?

Solution:

- The sample space is $S = \{HH, HT, TH, TT\}$, where H and T denote head and tail.
- Since we assume that the coin is balanced, these outcomes are equally likely and we assign to each sample point the probability $\frac{1}{4}$.
- Letting A denote the event that we will get at least one head, we get $A = \{HH, HT, TH\}$ and

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(HH) + \mathbb{P}(HT) + \mathbb{P}(TH) \\ &= \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}\end{aligned}$$

Exercise 1.

Suppose an experiment has an infinite sequence of outcomes $O_1, O_2, O_3, \dots$, such that

$$\mathbb{P}(O_i) = \left(\frac{1}{2}\right)^i \quad \text{for } i = 1, 2, \dots$$

is this a properly defined probability?

Solution 1.

- Note that $S = \{O_1, O_2, O_3, \dots\}$
- Since the probabilities are all positive, it remains to be shown that $\mathbb{P}(S) = 1$.
- Getting

$$\begin{aligned}\mathbb{P}(S) &= \mathbb{P}(O_1) + \mathbb{P}(O_2) + \mathbb{P}(O_3) + \dots \\ &= \left(\frac{1}{2}\right)^1 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \dots \\ &= \frac{\frac{1}{2}}{1 - \frac{1}{2}} \\ &= 1\end{aligned}$$

Theorem 2.

If an experiment can result in any one of N different equally likely outcomes, and if n of these outcomes together constitute event A , then the probability of event A is

$$\mathbb{P}(A) = \frac{n}{N}$$

Proof.

Let $O_1, O_2, O_3, \dots, O_n$ represent the individual outcomes in S , each with probability $\frac{1}{N}$. If A is the union of n of these mutually exclusive outcomes, and it does not matter which ones, then

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(O_1 \cup O_2 \cup \dots \cup O_n) \\ &= \mathbb{P}(O_1) + \mathbb{P}(O_2) + \dots + \mathbb{P}(O_n) \\ &= \underbrace{\frac{1}{N} + \frac{1}{N} + \dots + \frac{1}{N}}_{n \text{ terms}} = \frac{n}{N}.\end{aligned}$$



Example 6.

Toss two fair coins and define $A = \{\text{one head and one tail}\}$, calculate $\mathbb{P}(A)$.

Solution:

- The sample space is $S = \{HT, TH, HH, TT\}$
- $A = \{HT\} \cup \{TH\}$, thus, $\mathbb{P}(A) = \frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$.

Theorem 3 (Properties).

A probability $\mathbb{P}$ satisfies the following properties:

1. For each event A , $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
2. The probability of the null set is zero; that is, $\mathbb{P}(\emptyset) = 0$.
3. $\mathbb{P}(A) \leq \mathbb{P}(B)$ whenever $A \subset B$.
4. For each A , $0 \leq \mathbb{P}(A) \leq 1$.

Proof.

1. We have $S = A \cup A^c$ and $A \cap A^c = \emptyset$. Thus, from (2) and (3) of Definition (9), it follows that

$$1 = \mathbb{P}(A) + \mathbb{P}(A^c).$$

2. In the first Item, take $A = \emptyset$ so that $A^c = S$. Accordingly, we have

$$\mathbb{P}(\emptyset) = 1 - \mathbb{P}(S) = 0.$$

3. Now $B = A \cup (A^c \cap B)$ and $A \cap (A^c \cap B) = \emptyset$. Hence, from (3) of Definition (9),

$$\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(A^c \cap B).$$

From (1) of Definition (9), $\mathbb{P}(A^c \cap B) \geq 0$. Hence $\mathbb{P}(A) \leq \mathbb{P}(B)$.

4. Since $\emptyset \subset A \subset S$, we have by (3) that

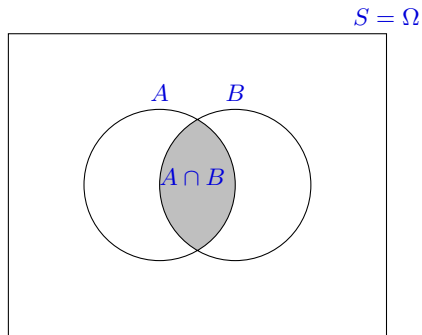
$$\mathbb{P}(\emptyset) \leq \mathbb{P}(A) \leq \mathbb{P}(S) \text{ or } 0 \leq \mathbb{P}(A) \leq 1.$$



Theorem 4.

- For any events A and B , we have

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$



Proof.

- Each of the sets $A \cup B$ and B can be represented, respectively, as a union of non intersecting sets as follows:

$$A \cup B = A \cup (A^c \cap B) \text{ and } B = (A \cap B) \cup (A^c \cap B).$$

Thus, from (3) of Definition (9),

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(A^c \cap B)$$

and

$$\mathbb{P}(B) = \mathbb{P}(A \cap B) + \mathbb{P}(A^c \cap B).$$

If the second of these equations is solved for $\mathbb{P}(A^c \cap B)$ and this result substituted in the first equation, we obtain

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

This completes the proof.



Example 7.

In a large metropolitan area, the probabilities are 0.86, 0.35, and 0.29, respectively, that a family (randomly chosen for a sample survey) owns a color television set, a HDTV set, or both kinds of sets. What is the probability that a family owns either or both kinds of sets?

Solution:

If A is the event that a family in this metropolitan area owns a color television set and B is the event that it owns a HDTV set, we have $\mathbb{P}(A) = 0.86$, $\mathbb{P}(B) = 0.35$, and $\mathbb{P}(A \cap B) = 0.29$; substitution into the formula of Theorem 4 yields

$$\mathbb{P}(A \cup B) = 0.86 + 0.35 - 0.29 = 0.92.$$

Example 8.

A population individuals can be cross-classified by gender and smoking habits as:

	Female	Male	Total
Smoker	0.2	0.25	0.45
non-smoker	0.31	0.24	0.55
Total	0.51	0.49	1

Let

- A = “ person is a male”
- B = “ person is non-smoker”

Calculate $P(A \cup B)$

Solution:

	Female	Male	Total
Smoker	$\mathbb{P}(A^c \cap B^c) = 0.2$	$\mathbb{P}(A \cap B^c) = 0.25$	$\mathbb{P}(B^c) = 0.45$
non-smoker	$\mathbb{P}(A^c \cap B) = 0.31$	$\mathbb{P}(A \cap B) = 0.49$	$\mathbb{P}(B) = 0.55$
Total	$\mathbb{P}(A^c) = 0.51$	$\mathbb{P}(A) = 0.49$	1

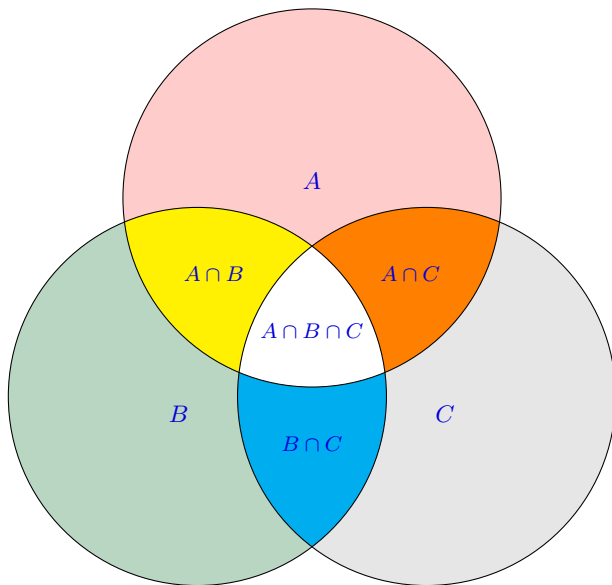
Using Theorem 4, we get

$$\begin{aligned}
 \mathbb{P}(A \cup B) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \\
 &= 0.49 + 0.55 - 0.24 \\
 &= 0.8
 \end{aligned}$$

Theorem 5.

- If A, B , and C are any three events in a sample space S , then

$$\begin{aligned}\mathbb{P}(A \cup B \cup C) = & \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) \\ & - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C).\end{aligned}$$



Proof of Theorem 5.

Writing $A \cup B \cup C$ as $A \cup (B \cup C)$ and using Theorem 4 twice, once for $\mathbb{P}[A \cup (B \cup C)]$ and once for $\mathbb{P}(B \cup C)$, we get

$$\begin{aligned}\mathbb{P}(A \cup B \cup C) &= \mathbb{P}[A \cup (B \cup C)] \\ &= \mathbb{P}(A) + \mathbb{P}(B \cup C) - \mathbb{P}[A \cap (B \cup C)] \\ &= \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(B \cap C) - \mathbb{P}[A \cap (B \cup C)]\end{aligned}$$

Then, using the distributive law, we find that

$$\begin{aligned}\mathbb{P}[A \cap (B \cup C)] &= \mathbb{P}[(A \cap B) \cup (A \cap C)] \\ &= \mathbb{P}(A \cap B) + \mathbb{P}(A \cap C) - \mathbb{P}[(A \cap B) \cap (A \cap C)] \\ &= \mathbb{P}(A \cap B) + \mathbb{P}(A \cap C) - \mathbb{P}(A \cap B \cap C)\end{aligned}$$

and hence

$$\begin{aligned}\mathbb{P}(A \cup B \cup C) &= \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) \\ &\quad - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C).\end{aligned}$$



Example 9.

If a person visits his dentist, suppose that the probability that he will have his teeth cleaned is 0.44, the probability that he will have a cavity filled is 0.24, the probability that he will have a tooth extracted is 0.21, the probability that he will have his teeth cleaned and a cavity filled is 0.08, the probability that he will have his teeth cleaned and a tooth extracted is 0.11, the probability that he will have a cavity filled and a tooth extracted is 0.07, and the probability that he will have his teeth cleaned, a cavity filled, and a tooth extracted is 0.03. What is the probability that a person visiting his dentist will have at least one of these things done to him?

Solution:

Define

- C is the event that the person will have his teeth cleaned
- F is the event that he will have a cavity filled
- E is the event that he will have a tooth extracted

we are given

- $\mathbb{P}(C) = 0.44, \mathbb{P}(F) = 0.24, \mathbb{P}(E) = 0.21,$
- $\mathbb{P}(C \cap F) = 0.08, \mathbb{P}(C \cap E) = 0.11, \mathbb{P}(F \cap E) = 0.07,$
- and $\mathbb{P}(C \cap F \cap E) = 0.03,$

Substitution into the formula of Theorem 5 yields

$$\begin{aligned}\mathbb{P}(C \cup F \cup E) &= 0.44 + 0.24 + 0.21 - 0.08 - 0.11 - 0.07 + 0.03 \\ &= 0.66.\end{aligned}$$

Exercise 2.

Ben is going to celebrate the beginning of the year of the dragon. He lives close to two pubs. The probability that he would go to pub A is 0.5 and the probability that he would go to pub B is 0.4 . In addition, the probability that he would go to at least one of the two venues is 0.8 .

1. What is the sample space?
2. What is the probability that he would go to both pubs?
3. What is the probability that he would go to exactly one pub?

Solution 2.

1. Let A be the event that he would go to pub A , and let B be the event that he goes to pub B . Therefore,

$$S = \{A \cap B^c, A^c \cap B, A \cap B, A^c \cap B^c\}$$

2. The probability that he would go to both pubs is

$$\begin{aligned}\mathbb{P}(A \cap B) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B) \\ &= 0.5 + 0.4 - 0.8 \\ &= 0.1\end{aligned}$$

3. The probability that he would go to exactly one pub is

$$\begin{aligned}\mathbb{P}\left((A \cup B) \setminus (A \cap B)\right) &= \mathbb{P}(A \cup B) - \mathbb{P}(A \cap B) \\ &= 0.8 - 0.1 \\ &= 0.7\end{aligned}$$

Exercise 3.

A lady crosses three traffic signals, with red and green lights only, on the way to her dog's hairdresser.

The probabilities of encountering 0, 1, and 2 red lights are 0.4, 0.1, 0.2 respectively.

Find the probabilities of

1. Encountering at least one red light.
2. Encountering at least one green light.
3. Encountering an odd number of red lights.

Solution 3.

1. The required probability is

$$\begin{aligned}\mathbb{P}(1 \text{ red}) + \mathbb{P}(2 \text{ red}) + \mathbb{P}(3 \text{ red}) &= 1 - \mathbb{P}(0 \text{ red}) \\ &= 1 - 0.4 \\ &= 0.6\end{aligned}$$

2. The required probability is

$$\begin{aligned}\mathbb{P}(1 \text{ green}) + \mathbb{P}(2 \text{ green}) + \mathbb{P}(3 \text{ green}) &= \mathbb{P}(0 \text{ red}) + \mathbb{P}(1 \text{ red}) + \mathbb{P}(2 \text{ red}) \\ &= 0.4 + 0.1 + 0.2 \\ &= 0.7\end{aligned}$$

3. The required probability is

$$\begin{aligned}\mathbb{P}(1 \text{ red}) + \mathbb{P}(3 \text{ red}) &= 1 - \mathbb{P}(0 \text{ red}) - \mathbb{P}(2 \text{ red}) \\ &= 1 - 0.4 - 0.2 \\ &= 0.4\end{aligned}$$

Example 10.

Take Example 8, but assume this time we have the numbers in millions

	Married (M)	Non-married (NM)	Total
Smoker (S)	20	25	45
non-smoker (N)	31	24	55
Total	51	49	100

- $\mathbb{P}(M) = \frac{51}{100} = 0.51$, etc, ...
- What is the probability of the event “a person is smoker *GIVEN* that they are not married”?

Solution:

- This event is conditional: “Smoker **given that** the person is non-married”
- We only look into non-married population

-

$$\mathbb{P}(S \mid NM) = \frac{25}{49} \approx 0.51.$$

Definition 10.

If A and B are any two events in a sample space S and $\mathbb{P}(A) > 0$, the conditional probability of B given A is

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}. \quad (1)$$

Example 11.

A manufacturer of airplane parts knows from past experience that the probability is 0.80 that an order will be ready for shipment on time, and it is 0.72 that an order will be ready for shipment on time and will also be delivered on time. What is the probability that such an order will be delivered on time given that it was ready for shipment on time?

Example 11.

A manufacturer of airplane parts knows from past experience that the probability is 0.80 that an order will be ready for shipment on time, and it is 0.72 that an order will be ready for shipment on time and will also be delivered on time. What is the probability that such an order will be delivered on time given that it was ready for shipment on time?

Solution:

If we let

- R stand for the event that an order is ready for shipment on time
- D be the event that it is delivered on time
- we have $\mathbb{P}(R) = 0.80$ and $\mathbb{P}(R \cap D) = 0.72$
- it follows that

$$\mathbb{P}(D \mid R) = \frac{\mathbb{P}(R \cap D)}{\mathbb{P}(R)} = \frac{0.72}{0.80} = 0.90.$$

- If we multiply the expressions on both sides of the formula of Definition 10 by $\mathbb{P}(A)$, we obtain the following **multiplication rule**.

$$\mathbb{P}(A) \cdot \mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \cap B) \cdot \cancel{\mathbb{P}(A)}}{\cancel{\mathbb{P}(A)}} = \mathbb{P}(A \cap B).$$

Theorem 6.

If A and B are any two events in a sample space S and $\mathbb{P}(A) \neq 0$, then

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B \mid A).$$

Example 12.

Find the probabilities of randomly drawing two aces in succession from an ordinary deck of 52 playing cards if we sample

- (a) without replacement;
- (b) with replacement.

Example 12.

Find the probabilities of randomly drawing two aces in succession from an ordinary deck of 52 playing cards if we sample

- (a) without replacement;
- (b) with replacement.

Solution:

- (a) If the first card is not replaced before the second card is drawn, the probability of getting two aces in succession is

$$\frac{4}{52} \times \frac{3}{51} = \frac{1}{221}$$

- (b) If the first card is replaced before the second card is drawn, the corresponding probability is

$$\frac{4}{52} \times \frac{4}{52} = \frac{1}{169}$$

Theorem 7.

If A, B , and C are any three events in a sample space S such that $\mathbb{P}(A \cap B) \neq 0$, then

$$\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A) \cdot \mathbb{P}(B | A) \cdot \mathbb{P}(C | A \cap B).$$

Proof of Theorem 7.

Writing $A \cap B \cap C$ as $(A \cap B) \cap C$ and using the formula of Theorem 6 twice, we get

$$\begin{aligned}\mathbb{P}(A \cap B \cap C) &= \mathbb{P}[(A \cap B) \cap C] \\ &= \mathbb{P}(A \cap B) \cdot \mathbb{P}(C | A \cap B) \\ &= \mathbb{P}(A) \cdot \mathbb{P}(B | A) \cdot \mathbb{P}(C | A \cap B).\end{aligned}$$



Example 13.

A box of fuses contains 20 fuses, of which 5 are defective. If 3 of the fuses are selected at random and removed from the box in succession without replacement, what is the probability that all 3 fuses are defective?

Example 13.

A box of fuses contains 20 fuses, of which 5 are defective. If 3 of the fuses are selected at random and removed from the box in succession without replacement, what is the probability that all 3 fuses are defective?

Solution:

Let

- A be the event that the first fuse is defective,
- B be the event that the second fuse is defective,
- and C be the event that the third fuse is defective,
- then

$$\mathbb{P}(A) = \frac{5}{20}, \mathbb{P}(B | A) = \frac{4}{19}, \mathbb{P}(C | A \cap B) = \frac{3}{18},$$

and using Theorem 7, we get

$$\begin{aligned}\mathbb{P}(A \cap B \cap C) &= \mathbb{P}(A) \cdot \mathbb{P}(B | A) \cdot \mathbb{P}(C | A \cap B) \\ &= \frac{5}{20} \cdot \frac{4}{19} \cdot \frac{3}{18} = \frac{1}{114}.\end{aligned}$$

Definition 11.

Two events A and B are independent if and only if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$$

Example 14.

A coin is tossed three times and the eight possible outcomes, $HHH, HHT, HTH, THH, HTT, THT, TTH$, and TTT , are assumed to be equally likely. If A is the event that a head occurs on each of the first two tosses, B is the event that a tail occurs on the third toss, and C is the event that exactly two tails occur in the three tosses, show that

- (a)- events A and B are independent;
- (b)- events B and C are dependent

Since

$$A = \{HHH, HHT\}$$

$$B = \{HHT, HTT, THT, TTT\}$$

$$A \cap B = \{HHT\}$$

$$B \cap C = \{HTT, THT\}.$$

The assumption that the eight possible outcomes are all equiprobable yields

$$\mathbb{P}(A) = \frac{1}{4}, \mathbb{P}(B) = \frac{1}{2}, \mathbb{P}(C) = \frac{3}{8}, \mathbb{P}(A \cap B) = \frac{1}{8}, \text{ and } \mathbb{P}(B \cap C) = \frac{1}{4}.$$

- (a)- Since $\mathbb{P}(A) \cdot \mathbb{P}(B) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8} = \mathbb{P}(A \cap B)$, events A and B are independent.
- (b)- Since $\mathbb{P}(B) \cdot \mathbb{P}(C) = \frac{1}{2} \cdot \frac{3}{8} = \frac{3}{16} \neq \mathbb{P}(B \cap C)$, events B and C are not independent.

Theorem 8.

If A and B are independent, then A and B^c are also independent.

Proof of Theorem 8.

Since $A = (A \cap B) \cup (A \cap B^c)$, $(A \cap B)$ and $(A \cap B^c)$ are mutually exclusive, and A and B are independent by assumption, we have

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}[(A \cap B) \cup (A \cap B^c)] \\ &= \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c) \quad (\text{Using Axiom 3}) \\ &= \mathbb{P}(A) \cdot \mathbb{P}(B) + \mathbb{P}(A \cap B^c) \quad (A \text{ and } B \text{ indep.})\end{aligned}$$

It follows that

$$\begin{aligned}\mathbb{P}(A \cap B^c) &= \mathbb{P}(A) - \mathbb{P}(A) \cdot \mathbb{P}(B) \\ &= \mathbb{P}(A) \cdot [1 - \mathbb{P}(B)] \\ &= \mathbb{P}(A) \cdot \mathbb{P}(B^c)\end{aligned}$$

and hence that A and B^c are independent. □

Definition 12.

Events $A_1, A_2, \dots$, and A_k are independent if and only if the probability of the intersections of any $2, 3, \dots$, or k of these events equals the product of their respective probabilities.

- For three events A, B and C , for example, independence requires that

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$$

$$\mathbb{P}(A \cap C) = \mathbb{P}(A) \cdot \mathbb{P}(C)$$

$$\mathbb{P}(B \cap C) = \mathbb{P}(B) \cdot \mathbb{P}(C)$$

$$\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A) \cdot \mathbb{P}(B) \cdot \mathbb{P}(C)$$

Exercise 4.

Find the probabilities of getting

- (a)- three heads in three random tosses of a balanced coin;
- (b)- four sixes and then another number in five random rolls of a balanced die.

Exercise 5.

Find the probabilities of getting

- (a)- three heads in three random tosses of a balanced coin;
- (b)- four sixes and then another number in five random rolls of a balanced die.

Solution 5.

- (a)- The probability of a head on each toss is $\frac{1}{2}$ and the three outcomes are independent. Thus we can multiply, obtaining

$$\frac{1}{2} \cdots \frac{1}{2} \cdots \frac{1}{2} = \frac{1}{8}$$

- (b)- The probability of a six on each toss is $\frac{1}{6}$; thus the probability of tossing a number other than 6 is $\frac{5}{6}$. Inasmuch as the tosses are independent, we can multiply the respective probabilities to obtain

$$\frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{5}{6} = \frac{5}{7776}$$

Theorem 9.

If the events $B_1, B_2, \dots$, and B_k constitute a partition of the sample space S and $\mathbb{P}(B_i) \neq 0$ for $i = 1, 2, \dots, k$, then for any event A in S

$$\mathbb{P}(A) = \sum_{i=1}^k \mathbb{P}(A \cap B_i) = \sum_{i=1}^k \mathbb{P}(B_i) \cdot \mathbb{P}(A \mid B_i)$$

- The B 's constitute a partition of the sample space if they are pairwise mutually exclusive and if their union equals S .

Proof.

We have $\cup_{i=1}^k B_i = S$, $B_i \cap B_j = \emptyset, \forall i \neq j$ and $(A \cap B_i) \cap (A \cap B_j) = \emptyset, \forall i \neq j$. Observe that

$$\begin{aligned} A &= A \cap S \\ &= A \cap (B_1 \cup B_2 \cup \dots \cup B_k) \\ &= (A \cap B_1) \cup (A \cap B_2) \cup \dots \cup (A \cap B_k) \end{aligned}$$

By (3) of Definition (9) and Theorem 6, we get

$$\mathbb{P}(A) = \mathbb{P}\left(\bigcup_{i=1}^k A \cap B_i\right) = \sum_{i=1}^k \mathbb{P}(A \cap B_i) = \sum_{i=1}^k \mathbb{P}(B_i) \cdot \mathbb{P}(A | B_i).$$



Example 15.

The members of a consulting firm rent cars from three rental agencies: 60 percent from agency 1, 30 percent from agency 2, and 10 percent from agency 3. If 9 percent of the cars from agency 1 need an oil change, 20 percent of the cars from agency 2 need an oil change, and 6 percent of the cars from agency 3 need an oil change, what is the probability that a rental car delivered to the firm will need an oil change?

Solution: Define the events:

- A = the car needs an oil change,
- B_1 , B_2 and B_3 the car comes from rental agencies 1, 2, or 3

We have $P(B_1) = 0.60$, $P(B_2) = 0.30$, $P(B_3) = 0.10$, $P(A | B_1) = 0.09$, $P(A | B_2) = 0.20$, and $P(A | B_3) = 0.06$. Substituting these values into the formula of Theorem 9, we get

$$\begin{aligned} P(A) &= P(B_1).P(A | B_1) + P(B_2).P(A | B_2) + P(B_3).P(A | B_3) \\ &= (0.60)(0.09) + (0.30)(0.20) + (0.10)(0.06) = 0.12 \end{aligned}$$

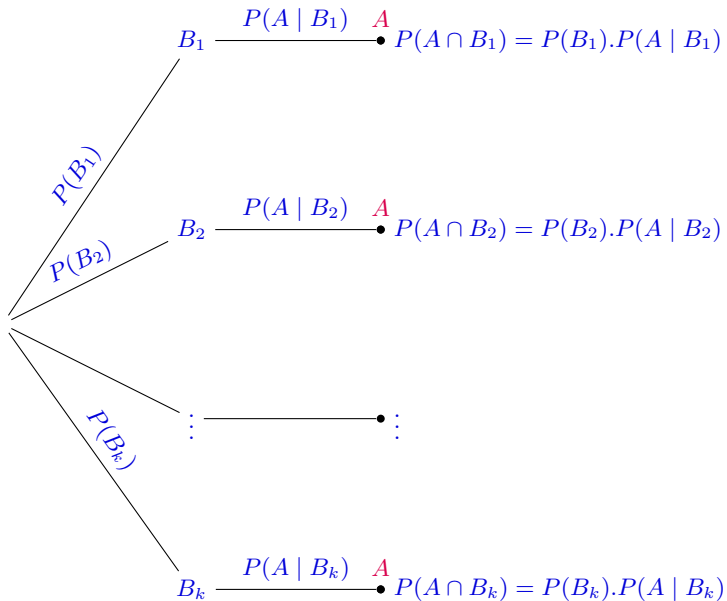
Theorem 10.

If the events $B_1, B_2, \dots$, and B_k constitute a partition of the sample space S and $\mathbb{P}(B_i) \neq 0$ for $i = 1, 2, \dots, k$, then for any event A in S , such that $\mathbb{P}(A) \neq 0$

$$\mathbb{P}(B_r | A) = \frac{\mathbb{P}(B_r) \cdot \mathbb{P}(A | B_r)}{\sum_{i=1}^k \mathbb{P}(B_i) \cdot \mathbb{P}(A | B_i)}$$

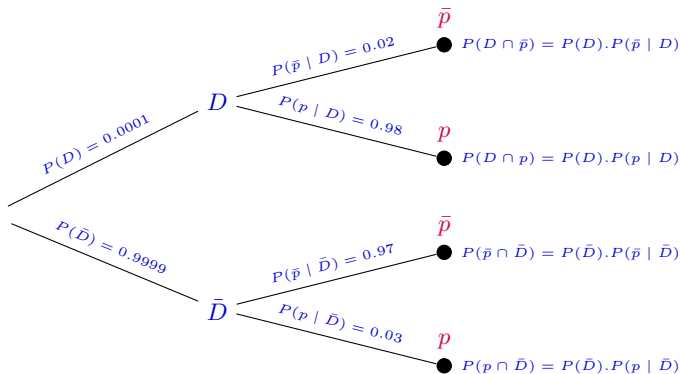
for $r = 1, 2, \dots, k$.

Proof: The proof will be given in class.



Example 16.

A rare but serious disease, D , has been found in 0.01 percent of a certain population. A test has been developed that will be positive, p , for 98 percent of those who have the disease and be positive for only 3 percent of those who do not have the disease. Find the probability that a person tested as positive does not have the disease.



- Define the events
 - $\bar{D}$ = “person does not have the disease”
 - $\bar{p}$ = “ person is found negative for the disease”
- Substituting the given probabilities into the formula of Theorem 10, we get

$$\begin{aligned}\mathbb{P}(\bar{D} | p) &= \frac{\mathbb{P}(\bar{D})\mathbb{P}(p | \bar{D})}{\mathbb{P}(D)\mathbb{P}(p | D) + \mathbb{P}(\bar{D})\mathbb{P}(p | \bar{D})} \\ &= \frac{0.9999 \times 0.03}{0.0001 \times 0.98 + 0.9999 \times 0.03} \\ &= 0.997\end{aligned}$$

- For your practice do the following exercises:
 1. Exercises 50, 51 and 52 page 53.
 2. Exercises 54 and 58 page 59.
 3. Exercises 76 and 77 page 94.

Main textbook:

- John E. Freund's. *Mathematical Statistics With Applications* Ed: 8 (Required).

Suggested textbooks:

- Ross, S., 2009. *A first course in probability*. Upper Saddle River, 6.
- Hossein Pishro-Nik. *Probability, Statistics and Random Processes* [here](#).

Some useful websites:

- [Society of Actuaries, Exam of probability, sample questions](#).
- [Society of Actuaries, Exam of probability, sample solutions](#).
- Some contents of set theory were inspired from [here](#).